Digital Payments Settlement Latency & Merchant Churn Analytics

Fix breach-exposed SMB settlement and finish the smart-routing rollout to protect ₹11.6M GMV and ₹151K MDR a year

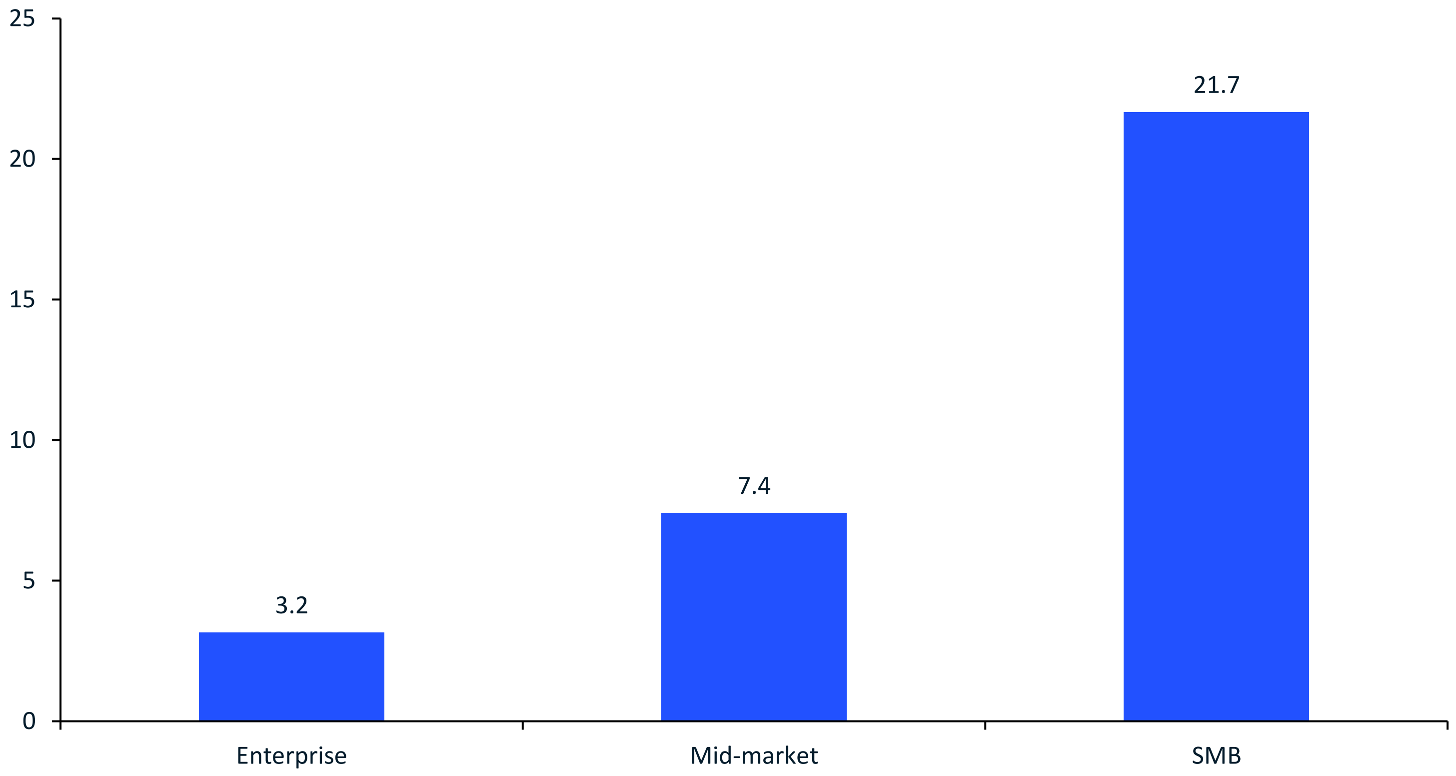
- **Two failure modes, one lever:** PSU switch latency and T+2/T+3 settlement breaches drive SMB churn — and SMB pays 65% of platform MDR.
- **The engine works:** Dynamic SLA routing already delivers +8.57 pp success (79.4% → 88.0%, benchmark band 8.30–8.70 ✓) and cuts latency 39% (1,700 → 1,035 ms).
- **The prize:** 894 at-risk SMB merchants (>25% of batches breaching) carry ₹51.6M annualized GMV / ₹0.67M MDR; fixing breaches protects ≈ ₹11.6M GMV / ₹151K MDR per year.
- **The ask:** fund the T+1 settlement guarantee for breach-exposed SMB, complete routing coverage to 100% of traffic, and stand up a save-desk at the 20–25% breach trigger.

Source: payments_analytics.db — 530,000 txns, 125,000 settlements, 2023-07→2024-12 (Group 03 analysis).

SITUATION

Churn concentrates in SMB — 21.7% vs 3.2% for enterprise — and half of SMB churners blame settlement delay

- **Scale:** 530,000 transactions across 8,500 merchants; 16.21% churn overall in 18 months.
- **Churn reasons (SMB):** SETTLEMENT_DELAY 581 merchants (49% of SMB churn) — ahead of high failure rate (281) and competitor switches (227).
- **Why it matters:** SMB delivers ₹1.85M of ₹2.83M realized MDR (65%) — SMB attrition is a direct P&L event.

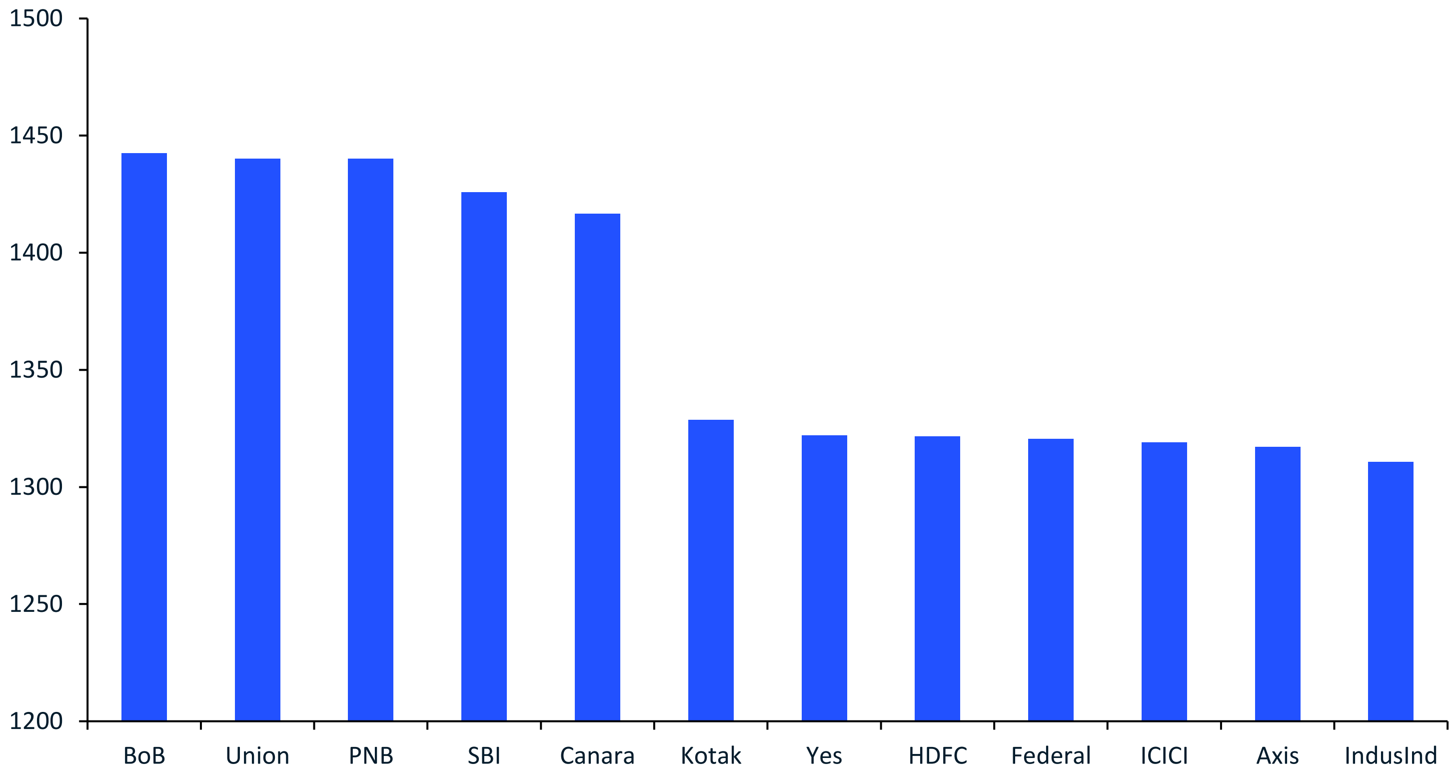


Churn rate (% of merchants)

Source: dim_merchants, fact_merchant_settlements — 8,500 merchants, 18 months.

All five hottest switches are PSU banks running ~100 ms hotter than private peers — an addressable routing problem

- **Tier gap:** PSU banks average 1,433 ms vs 1,322 ms for Tier-1 private (+8%); p95 spans ~7.0 s (PSU) vs ~4.1 s (private); spike share (≥10 s) 1.92% vs 1.47%.
- **Peak stress:** 18:00–22:00 latency 1,541 ms vs 1,321 ms off-peak; spikes ~508/hr vs ~329/hr (+55%) — success stays flat (83.3–84.1%), so the damage is experience-level.
- **Implication:** the worst offenders are known, named, and concentrated — routing and SLOs, not capex, are the first move.

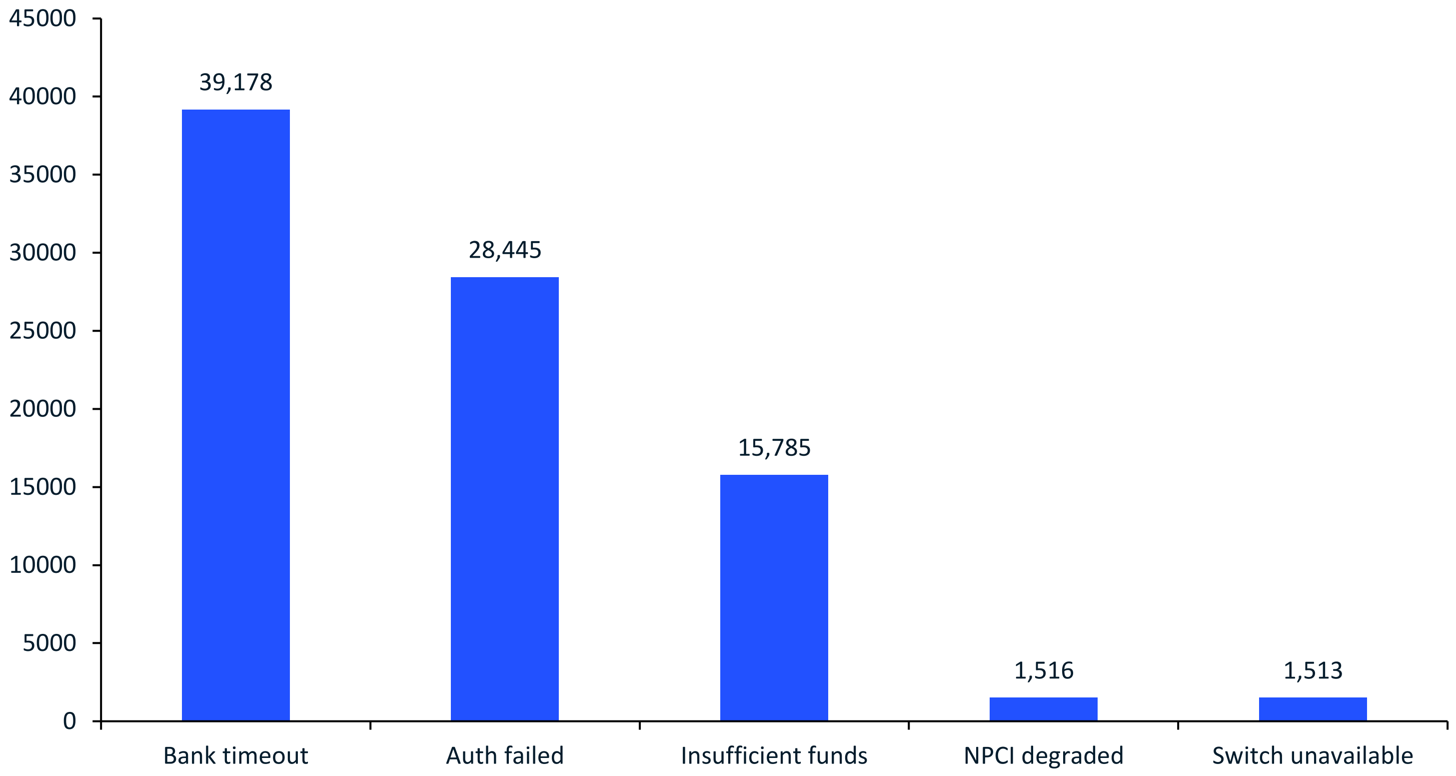


Average latency (ms)

Source: fact_payment_transactions × dim_issuing_banks — 530,000 attempts, 12 banks.

Nearly half of all 86,437 failures are infrastructure-side, led by bank timeouts at 45%

- **Failure load:** 86,437 failed attempts = 16.31% of all attempts.
- **Ownership split:** upstream infrastructure (bank timeout + NPCI degraded + switch unavailable) = 48.83%; user-side (auth + funds) = 51.17%.
- **Accounting note:** status TIMEOUT rows (21,611) and ERR_BANK_TIMEOUT-on-FAILED rows (17,567) are distinct populations — never summed.



Failed attempts (count)

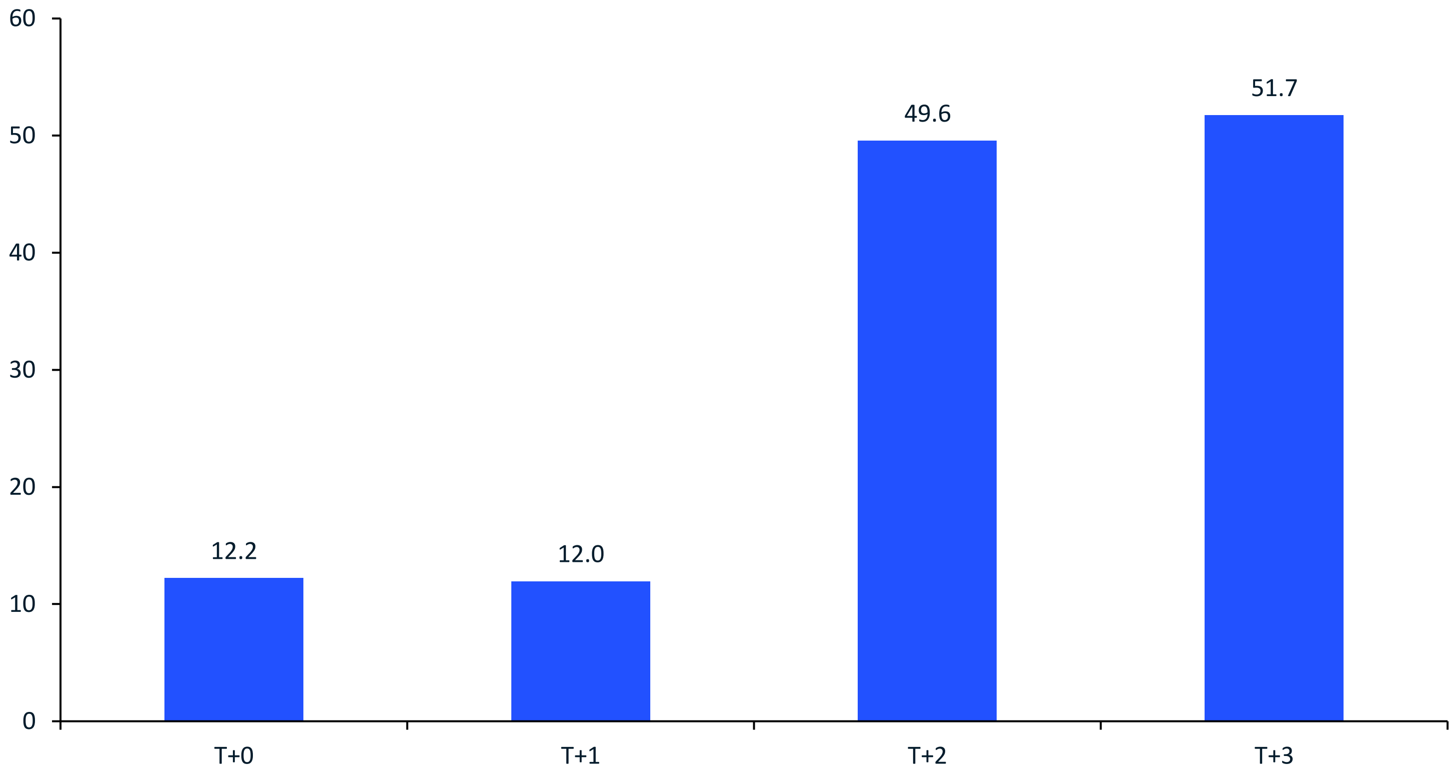
Source: fact_payment_transactions error codes, non-SUCCESS attempts (n=86,437).

T+2/T+3 contracts breach at ~50% — 4× the T+0/T+1 rate — with no seasonal relief all year

– **Structural, not noise:** long-cycle contracts breach at ~50% vs ~12% for T+0/T+1; breached batches still arrive ~1.5 days late.

– **Flat through time:** monthly breach rate holds 27.7–29.5% across 18 months — no seasonality, so run-rate math is legitimate.

– **Platform-wide:** the MCC × cycle matrix shows elevated T+2/T+3 breach in every merchant category — the batch pipeline, not a vertical, is failing.



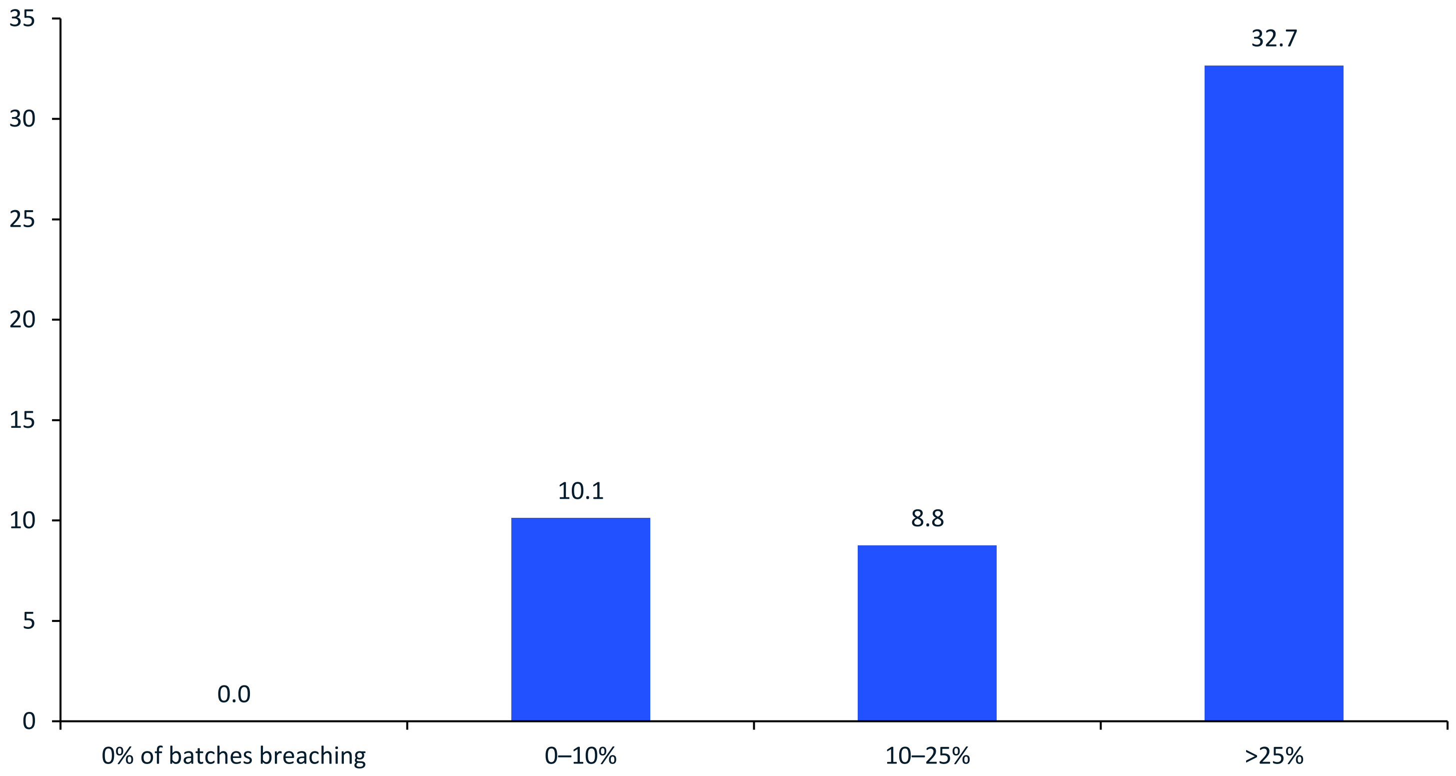
SLA breach rate (% of batches)

Above 25% breach exposure SMB churn triples to 32.7% — the save-trigger belongs at 20–25%

– **The cliff:** SMB churn in the >25% bucket runs 3.2× the low-breach baseline (32.66% vs 10.12%); mid-market shows the same direction (14.43%).

– **Mechanism confirmed:** churn is flat-to-down through moderate exposure, then inflects violently past 25% — settlement friction, not general dissatisfaction.

– **Operational trigger:** a rolling 30-day breach share crossing 20–25% is a warm, addressable churn signal — today nothing acts on it.

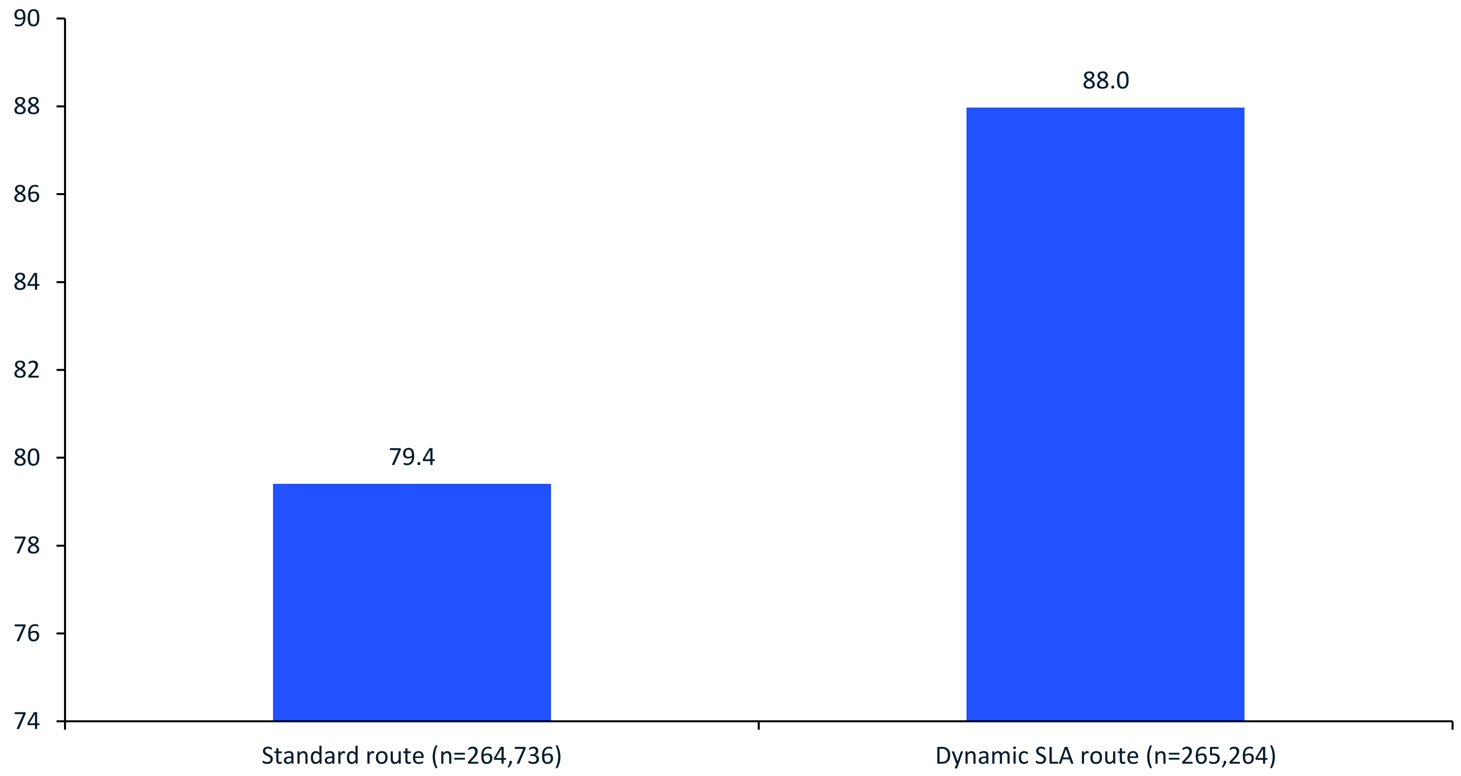


SMB churn rate (%)

Source: 2,517 settled merchants × breach exposure buckets (covered vs uncovered churn: 16.09% vs 16.26% — no selection bias).

Dynamic SLA routing already delivers +8.57 pp success (79.4% → 88.0%) and 39% lower latency — inside benchmark

- **Benchmark verified:** +8.57 pp absolute lift lands inside the accepted 8.30–8.70 band (target +8.50); stable by tier (8.21–8.74 pp) and channel (8.04–9.18 pp).
- **Latency dividend:** average 1,700 → 1,035 ms (−39%); p95 7,336 → 3,793 ms; ~666 ms clawed back on every degraded PSU switch.
- **Headroom:** 264,736 attempts still ride the standard route — each carries ~8.6 pp of avoidable failure risk.

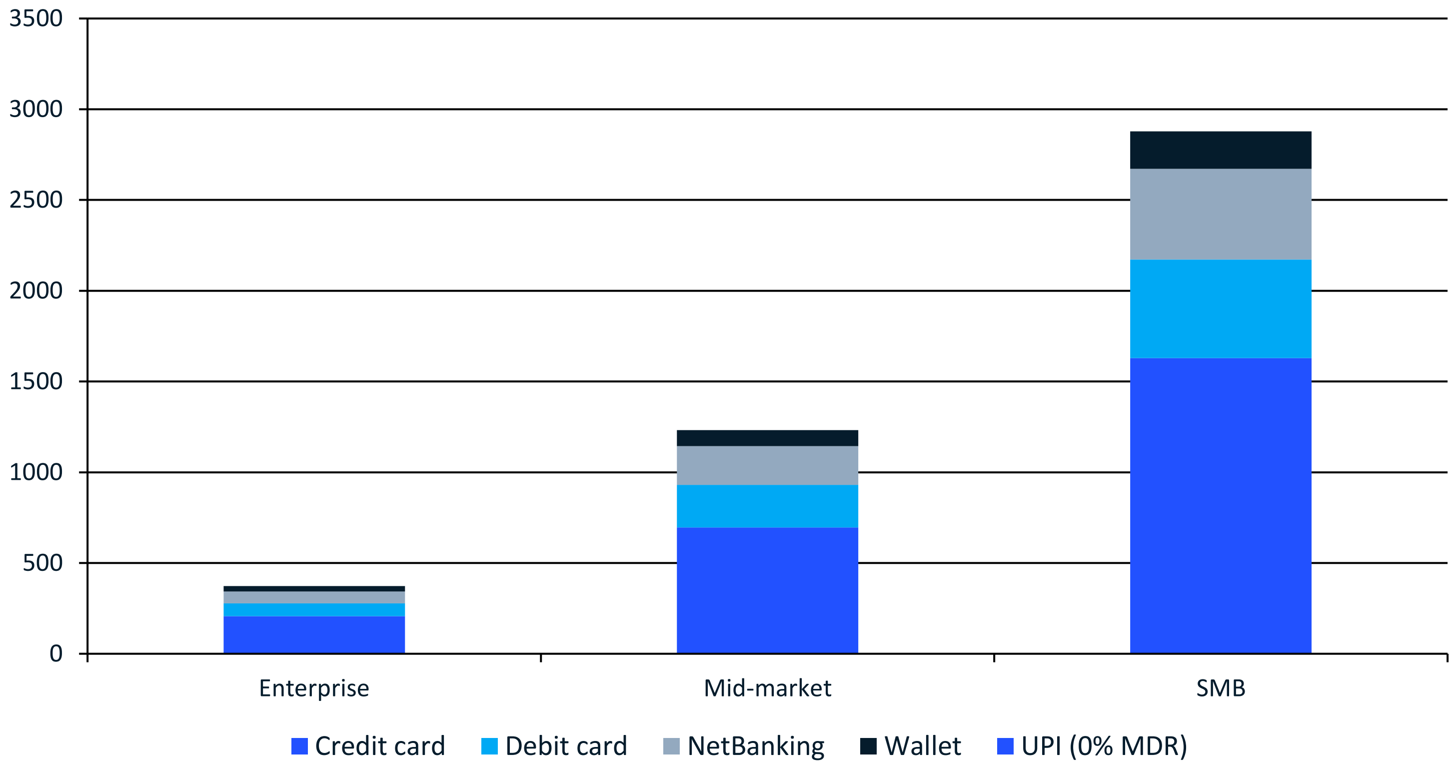


Success rate (%)

Source: fact_payment_transactions A/B split on is_routed_via_dynamic_sla (n=530,000).

SMB pays 65% of MDR while UPI carries 55% of volume free — and the at-risk SMB pool is ₹51.6M GMV a year

- **Monetization is card-shaped:** credit cards yield ≈56% of potential MDR from ~20% of successful GMV; UPI is 55% of GMV at a 0% rate.
- **At-risk pool:** 894 SMB merchants with >25% breach exposure = ₹77.4M GPV / ₹1.0M MDR over 18 months → ₹51.6M / ₹0.67M annualized.
- **Protected:** excess-churn model (32.66% → 10.12% baseline, ≈202 merchants) preserves ≈ ₹11.6M GMV / ₹151K MDR per year — assumptions stated in the notebook (run-rate annualization, churn reversion, no replacement credit).



Potential MDR (₹ '000, 18 mo)

Source: fact_merchant_settlements (realized ₹2.83M / ₹217.7M GPV) × transactions × mdr_rate_pct (potential ≈ ₹4.65M).

RECOMMENDATIONS

Five moves, led by the T+1 guarantee and full routing rollout, protect ₹11.6M GMV a year

- ★ 1. Guarantee T+1 settlement for breach-exposed SMB (liquidity buffer; halt new SMB T+2/T+3 contracts): protects ≈ ₹11.6M GMV / ₹151K MDR per year.
- ★ 2. Route 100% of traffic via Dynamic SLA; throttle evenings toward healthy rails; switch SLA scorecard with worst-3 renegotiation: ≈ 22.7k recovered transactions / 18 mo ≈ ₹36M GMV.
- ★ 3. Trigger a save-desk at a 20–25% rolling breach share (fee holiday, priority settlement): ≈ ₹2.9M GMV / ₹38K MDR per year at one-quarter capture.
- 4. Monetize the UPI base via value-added rails — instant-settlement SaaS at ₹1,000/mo, 10% attach on 5,455 SMB: ≈ ₹6.5M/yr, no transaction-fee change.
- 5. Enforce per-switch evening SLOs (p95 < 3 s, pre-scale 17:00–23:00, automated shedding): ≈ ₹0.1M GMV/month protected at peak.
- **Next step:** stand up the breach-exposure register from settlement data this week — every trigger in recommendations 1–3 keys off it.

Impacts quantified from this analysis; assumptions documented in Group_03_Digital_Payments_Settlement_Latency_SOLUTION.ipynb (KBQ 8).